

III.6 Degenerovaný Fermiho plyn

→ nerelativistické neinteragující fermiony (vodivostní pásy v kovech)

• silně kvantový režim $\beta\Lambda^3 \gtrsim 1$

• spin $1/2 \Rightarrow$ degenerace 2

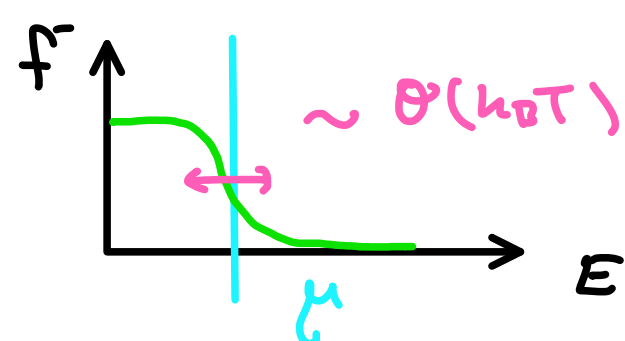
• hustota: $\rho = \frac{1}{V} \sum_k f_k^- \rightarrow \frac{1}{V} \int dE \mathcal{N}(E) f^-(E)$

$$\xrightarrow{V \rightarrow \infty} \Lambda^{-3}(T) \underbrace{g_{3/2}^-(2)}$$

konverguje pro $|z| \leq 1$ $= \sum_{n=1}^{\infty} (-1)^{n-1} \frac{z^n}{n^{3/2}}$

• ale: $\mu > 0$ je fyz. možné $\Rightarrow z > 1$ je možné

$\Rightarrow$ suma nekonverguje, je nutné použít **Sommerfeldův rozvoj kolem $T=0$**



→ Fermi-Diracova statistika:

$$f^-(E) = \frac{1}{1 + e^{\beta(E - \mu)}} = \frac{1}{2} - \frac{1}{2} \tanh \frac{E - \mu}{2k_B T} \xrightarrow{T \rightarrow 0^+} \Theta(\mu - E)$$

$T=0$

→ jsou zaplněny všechny stavy do μ

$$\rho = \frac{1}{V} \int_0^{\mu} \mathcal{N}(E) dE = \frac{1}{V} \underbrace{\Gamma(\mu)}$$

$$V \cdot g \cdot \frac{1}{h^3} \left. \frac{4}{3} \pi (2mE)^{3/2} \right|_{E=\mu}$$

→ Fermiho energie: $E_F = \mu|_{T=0}$

$$E_F = \frac{\hbar^2}{2m} (3\pi\rho)^{2/3} \propto \rho^{2/3}$$

• Fermiho teplota: $T_F = E_F / k_B$

• Fermiho hybnost $p_F = \hbar (3\pi\rho)^{1/3}$

→ tlak: $p = \frac{1}{V} \int_0^{\mu} \Gamma(E) dE = \frac{2}{5} \rho E_F \propto \rho^{5/2}$

→ hustota energie $u = \frac{3}{2} p = \frac{3}{5} \rho E_F \propto \rho^{5/2}$

$k_B T \ll E_F$

→ obecná situace:

$$I = \int_0^{\infty} \phi(E) f^-(E) dE \stackrel{PP}{=} \underbrace{\left[\mathcal{Z} f^- \right]_0^{\infty}}_0 - \int_0^{\infty} dE \mathcal{Z} \frac{\partial}{\partial E} \left[\frac{1}{2} - \frac{1}{2} \tanh \frac{E - \mu}{2k_B T} \right]$$

$$\uparrow$$

$$\phi(E) = \frac{d\mathcal{Z}}{dE}$$

$$\mathcal{Z}(0) = 0$$

$$= \int_0^{\infty} \mathcal{Z}(E) \frac{1}{4} \beta \frac{1}{\cosh^2 \frac{\beta(E - \mu)}{2}} dE$$

$\uparrow$ Taylor kolem $E = \mu$

$$= \frac{\beta}{4} \sum_{n=0}^{\infty} \mathcal{Z}^{(n)}(\mu) \int_0^{\infty} \frac{(E - \mu)^n}{n!} \frac{1}{\cosh^2 \frac{\beta(E - \mu)}{2}} dE$$

$$= \sum_{n=0}^{\infty} \frac{1}{4\beta^n} \mathcal{Z}^{(n)}(\mu) \int_{-\beta\mu}^{\infty} \frac{x^n}{n!} \frac{1}{\cosh^2 \frac{x}{2}} dx$$

$$\approx \sum_{n=0}^{\infty} \frac{\mathcal{Z}^{(n)}(\mu)}{n!} (k_B T)^n \underbrace{\int_{-\infty}^{\infty} \frac{x^n}{4 \cosh^2 \frac{x}{2}} dx}_{=0 \text{ pro liché } n, =1 \text{ pro } n=1}$$

$$\int_0^{\infty} \phi(E) f^-(E) dE = \int_0^{\mu} \phi(E) dE + \sum_{n=1}^{\infty} \frac{J_n}{(2n)!} \phi^{(2n-1)}(\mu) (k_B T)^{2n}$$

• speciálně $J_1 = \frac{\pi^2}{3}$

Korekce nejnižšího řádu

$$\rho = \frac{1}{V} \int_0^{\infty} \mathcal{N}(E) f^-(E) dE = \frac{1}{V} \int_0^{\mu} \mathcal{N}(E) dE + \frac{\pi^2}{6V} \underbrace{\mathcal{N}'(\mu)}_{\Gamma''(\mu)} (k_B T)^2$$

$$\frac{1}{V} \Gamma(\mu) = A \cdot \mu^{3/2} = \rho \left(\frac{\mu}{E_F} \right)^{3/2} \quad \Gamma''(\mu) = \frac{3}{2} \cdot \frac{1}{2} A V \mu^{-1/2} = \frac{3}{2} \cdot \frac{1}{2} A V \mu^{-1/2}$$

$$= \rho \left(\frac{\mu}{E_F} \right)^{3/2} + \frac{\pi^2}{8} \left(\frac{\mu}{E_F} \right)^{1/2} \left(\frac{k_B T}{E_F} \right)^2$$

vyřadíme $\frac{\mu}{E_F} = 1 - \frac{\pi^2}{12} \left(\frac{k_B T}{E_F} \right)^2 + O\left(\left(\frac{k_B T}{E_F} \right)^4 \right)$

→ tlak

$$p = \frac{1}{V} \int_0^{\infty} \Gamma(E) f^-(E) dE = \frac{1}{V} \int_0^{\mu} \underbrace{\Gamma(E)}_{A \cdot V \cdot E^{3/2}} dE + \frac{\pi^2}{6} \frac{1}{V} \underbrace{\mathcal{N}'(\mu)}_{A \cdot V \cdot \frac{3}{2} \mu^{1/2}} (k_B T)^2$$

$$= \frac{2}{5} \rho E_F \left(\frac{\mu}{E_F} \right)^{5/2} + \frac{\pi^2}{4} \rho \left(\frac{\mu}{E_F} \right)^{1/2} \left(\frac{k_B T}{E_F} \right)^2 E_F$$

$$= \frac{2}{5} \rho E_F \left[\underbrace{\left(\frac{\mu}{E_F} \right)^{5/2}}_{1 - \frac{5}{2} \frac{\pi^2}{12} \left(\frac{k_B T}{E_F} \right)^2} + \frac{5}{8} \pi^2 \underbrace{\left(\frac{\mu}{E_F} \right)^{1/2}}_{\approx 1} \left(\frac{k_B T}{E_F} \right)^2 \right]$$

$$p = \frac{2}{5} \rho E_F \left[1 + \frac{5\pi^2}{12} \left(\frac{k_B T}{E_F} \right)^2 \right]$$

$$\rightarrow u = \frac{3}{2} p = \frac{3}{5} \rho E_F \left[1 + \frac{5\pi^2}{12} \left(\frac{k_B T}{E_F} \right)^2 \right]$$

$$c_v = \frac{\partial u}{\partial T} = \frac{1}{2} k_B \rho \frac{k_B T}{E_F} \propto T$$